MVP

Minimum Viable Product Plan

PIL V1 — Core ML & API Layer

March 2026 | 12-Week Build Plan | Confidential

Document	Version	Status	Owner
PIL MVP Plan	1.0	Draft — In Review	Founding Team

1. What We Are Building

The PIL V1 MVP is the smallest possible version of the Prompt Intelligence Layer that proves the core value proposition: structured, compressed prompts produce equivalent or better LLM output at significantly lower token cost. V1 is backend-only — an API proxy that any developer can point their existing LLM API calls at. No IDE, no browser extension, no GUI beyond a minimal savings dashboard.

Principle	Ship the thing that proves token compression works and saves money. Everything else is V2.

1.1 MVP Must-Haves (Non-negotiable)

- PIL proxy API: drop-in replacement for OpenAI / Anthropic / Gemini API endpoints
- Intent classifier: classifies prompt task type (generate / debug / refactor / explain / document / test)
- Token budget engine: compresses prompts to a configurable token budget
- PII and secrets scrubber: removes entities and credentials before any LLM call
- Structured audit log: per-request metadata log with token delta and entity types found
- Savings summary endpoint: GET /stats returns aggregate tokens saved, cost saved, requests processed
- Docker image: single docker run command deploys the full stack locally or to any cloud
- Python SDK: pip install pil-sdk, three-line integration

1.2 MVP Explicitly NOT Included

- Rolling context summariser (deferred to V1 Beta)
- Semantic cache (deferred to V1 Beta)
- VS Code / JetBrains extension (V2)
- Web dashboard / front-end UI
- Team or org features
- Fine-tuning on customer data

2. Success Definition

The MVP is considered successful when all four of these conditions are met simultaneously within 60 days of first alpha user onboarding:

Condition	Metric	Measurement
Token reduction works	≥ 30% average token reduction on benchmark suite of 500 prompts	Automated eval pipeline
Quality is preserved	≥ 4.0 / 5.0 human eval score on compressed vs original outputs	10 design partners blind eval
Latency is acceptable	< 80ms p95 overhead added by PIL pipeline	Prometheus histogram
Users stick around	≥ 60% of alpha users make a second API call within 7 days	API request log retention metric

3. Build Plan — 12 Weeks

The 12-week build is divided into three 4-week sprints. Each sprint has a single primary goal and a hard done-when condition. If the done-when condition is not met, the sprint does not close.

Sprint 1 — Foundation (Weeks 1–4)

Goal: A working PIL proxy that intercepts prompts, scrubs PII, and returns cleaned prompts to the LLM. No compression yet. Prove the plumbing.

Sprint 1 — Foundation
Proxy works end-to-end. PII scrubber live. Audit log writes.

Task	Owner	Days
FastAPI proxy skeleton: routes, error handling, provider adapters (OpenAI, Anthropic)	Backend Eng	5 days
tiktoken + Anthropic tokeniser integration, token count headers on every response	Backend Eng	2 days
Microsoft Presidio integration: PII detection + placeholder substitution	ML Eng	4 days
Secret scanning: regex patterns for 15 common credential formats	ML Eng	2 days
Structured audit log: jsonl, per-request, local filesystem and optional S3	Backend Eng	3 days
Docker image + docker-compose with Redis	DevOps	2 days
Unit tests: 80% coverage on proxy routes and scrubber	All	2 days

Done when: Send real OpenAI API call through PIL proxy. PII in prompt is replaced. Audit log entry written. Response headers include X-PIL-Tokens-Saved.

Sprint 2 — Compression (Weeks 5–8)

Goal: The intent classifier and token budget engine are live. Prompts come out measurably shorter and structurally better. Design partners can benchmark against their real workloads.

Sprint 2 — Compression
Token compression working. 30% reduction on benchmark. Design partners testing.

Task	Owner	Days
distilBERT intent classifier: training data prep, fine-tune, ONNX export	ML Eng	6 days

Sprint 2 — Compression Token compression working. 30% reduction on benchmark. Design partners testing.		
Prompt template library: 6 intent types × 3 provider formats = 18 templates	ML Eng	4 days
Token budget engine: TF-IDF scorer, pruner, budget modes (strict / adaptive / economy)	ML Eng	5 days
Benchmark eval pipeline: 500 prompt dataset, automated token delta + quality scoring	ML Eng	3 days
Python SDK: pip install pil-sdk, PILClient class, context manager, async support	Backend Eng	3 days
Alpha onboarding flow: API key generation, quickstart docs, Slack channel	Product	2 days
Onboard 10 design partners	Product	ongoing
Done when: Benchmark pipeline shows ≥ 30% average token reduction. 10 design partners have made at least 1 real API call through PIL.		

Sprint 3 — Harden & Launch (Weeks 9–12)

Goal: Production-ready. Performance validated. 50 design partners active. V1 alpha publicly announced.

Sprint 3 — Harden & Launch Production hardened. 50 design partners. Public alpha announced.		
Task	**Owner**	**Days**
Performance hardening: async pipeline, connection pooling, p95 latency < 80ms validated	Backend Eng	4 days
GET /stats savings endpoint: aggregate and per-user token/cost savings	Backend Eng	2 days
Circuit breaker: PIL bypasses itself if internal latency exceeds 200ms threshold	Backend Eng	2 days
Prometheus metrics + Grafana dashboard: latency, token delta, cache miss rate, error rate	DevOps	3 days
Security hardening: rate limiting, API key rotation, input size limits	Backend Eng	2 days
Human eval: 10 design partners blind-score compressed vs original outputs (100 pairs each)	Product / ML	5 days

Sprint 3 — Harden & Launch Production hardened. 50 design partners. Public alpha announced.		
Documentation: API reference, SDK guide, provider setup guides, privacy policy	Product	4 days
Onboard 40 more design partners. Public alpha announcement.	Product	ongoing
Done when: p95 latency < 80ms confirmed. Human eval ≥ 4.0/5.0. 50 design partners active. Docs published. Alpha announcement live.		

4. Team & Resources

Role	Headcount	Primary Responsibility	Sprint Focus
ML Engineer	1 (founder or hire)	Classifier, budget engine, eval pipeline	Sprints 1–3
Backend Engineer	1 (founder or hire)	Proxy, SDK, infrastructure, performance	Sprints 1–3
DevOps / Infra	0.5 (shared)	Docker, Kubernetes, Prometheus, CI/CD	Sprint 1, 3
Product / GTM	1 (founder)	Design partners, docs, announcement, feedback	Sprints 2–3

4.1 Infrastructure Cost — MVP Phase

Resource	Spec	Monthly Cost (est.)
API proxy (2× instances)	2 vCPU, 4GB RAM each — AWS t3.medium	~$70
ML inference (CPU only)	4 vCPU, 8GB RAM — AWS c5.xlarge	~$125
Redis (ElastiCache)	cache.t3.micro	~$20
Qdrant (future — V1 Beta)	Not in MVP scope	—
Monitoring (Grafana Cloud)	Free tier	$0
Total MVP infra		~$215 / month

5. Go-To-Market — Alpha

5.1 Design Partner Acquisition

The 50 design partners are the entire GTM focus for V1. No paid acquisition. No broad marketing. Targeted outreach only.

1. HackerNews Show HN post — 'I built a layer that compresses your LLM prompts and saves you money'
2. Direct DMs to developers publicly complaining about OpenAI bills on Twitter/X
3. Dev Discord communities: Latent Space, AI Engineers, HuggingFace, LangChain Discord
4. LinkedIn post targeting senior engineers and ML leads at Series A–C startups
5. Personal network and warm intros from advisors

5.2 Pricing — Alpha

Free Alpha	All features free for design partners during the 12-week alpha. No credit card. No rate limits (within reason). In exchange: feedback calls, benchmark data, testimonials.

Post-alpha pricing model (V1 GA):

- Free tier: 100,000 tokens processed/month through PIL. No cost savings sharing.
- Pro tier: $29/month. Unlimited requests. Savings dashboard. Priority support.
- Enterprise: Custom pricing. On-prem deployment. SOC 2. SLA. Audit log export.

5.3 Feedback Loop

Every design partner gets a Slack Connect channel. Weekly 30-minute feedback call for the first month. Monthly after that. Feedback is triaged weekly into: fix this sprint, fix next sprint, V2 backlog, won't fix.

6. Risks — MVP Specific

Risk	If It Happens	Response
< 30% token reduction on real prompts	Design partners see no ROI	Ship prompt diff view so they see what PIL changes; tune scorer; offer manual template mode
Latency > 80ms p95	Adds friction vs direct LLM call	Circuit breaker bypasses PIL; async pipeline; profile and optimise hot path
Design partners churn after one call	No feedback loop; no testimonials	Personal outreach within 24hr of first call; offer to co-author case study
PII scrubber false positive removes critical code	User loses context, output quality drops	User can mark code blocks as safe zones; review flagged entities before sending

PIL Product Team | MVP Plan v1.0 | March 2026